THE MASTER DOCUMENT · CONFIDENTIAL · WORKING REFERENCE V2

Classess Learner

The one document we return to when we are lost. Everything decided — the product, the psychology, the growth engine, the foundation, and the order of the build — with the reasoning attached, so no decision has to be re-made and nothing is forgotten.

D2C · AI-NATIVE

COGNITIVE FITNESS

GRADE 6 & UP · INDIA BOARDS FIRST

BRANDING & GROWTH OVER PROFIT

PRODUCTION-READY · NO MVP

COMPLIANT BY DESIGN

This is a living reference, not a pitch. It captures what we have converged on together. Where a thing is still open, it is marked as open. Where a thing is locked, the reasoning is recorded beside it — because the why is what keeps the what from drifting.

§ 00

Contents

PART I

The Thesis

The altitude everything else hangs from. What the product is, who it is for, the posture we are taking, and the laws that settle any decision the rest of the document does not.

§ 01

What we are building

Classess Learner is a direct-to-consumer, AI-native learning application for students — learners only, no teachers and no institutions inside it. It is the flagship product of the Dot eVentures education ecosystem, and it is built deliberately as the reference implementation: the first product that proves the pattern every future product will follow.

The altitude — read this first

The entity we are building is the **ecosystem**, not any single app. Its citizens are **KGtoPG** — the governed identity, data, and intelligence platform that every product plugs into and that belongs to no single app — and the products that sit on top of it, of which Learner is the first and the flagship. We architect ecosystem-first: the canonical identity, the academic ontology, the platform contracts, and the governance are designed as truths that hold across all products. Then Learner is built to plug into them correctly. **KGtoPG is a product in its own right, never Learner's database.** Learner is the proof that a product can plug into the platform cleanly — get that right once, and every future venture inherits identity, profile, and intelligence on day one.

The north star

Brilliant.org, but AI-native, built for India, premium, with mechanics nobody in edtech has shipped. We take the course structure from Khan Academy — board, grade, subject, chapters, topics, mastery — and the interactive tutoring vibe from Brilliant, both modified to our needs, not copied. The result is a personalised learning plan that actually produces results, wrapped in a habit loop that makes the app a daily ritual.

The reframe that governs everything — cognitive fitness

We are not "edtech." We are **cognitive fitness — a gym for the mind.** This is not a tagline; it is the decision that settles a dozen downstream choices. A gym is a ritual, not a cram tool. It is aspirational, not remedial. It is premium without apology. It is identity-forming — "I did my Classess" sits in the same mental slot as "I went to the gym." The daily free limit becomes a workout, not a paywall. Every time a design decision is unclear, ask what a great gym would do, not what an edtech app would do.

LOCKED POSTURE · THIS TURN

Branding and growth over profit, right now. We are not optimising revenue at this stage; we are growing reach and brand. This is a real strategic stance, and it threads through the whole document: generous free tier, aggressive growth loops, signature-quality everywhere ("spend whatever is needed"), and monetisation that is present and correct but deliberately not yet tuned for extraction. We buy reach with generosity, not with a cheap price.

Who it is for

The launch audience is **school-level learners, grade 6 (or each board's equivalent) and above**, across CBSE, ICSE, IB, Cambridge, state boards, and others — boards use different terminology and number their grades differently, so the model is "all of schooling from the middle-school level up," not a fixed age. We start with the **Indian academic boards** because that is our home market and we already understand it. There is no hard age bar; school level is the frame. The proving vertical for the first build is **middle-school mathematics** — chosen because symbolic verification makes correctness deterministic, misconception work is vivid, the visual stack is built for it, the prerequisite chain is well-mapped, and it is the highest-anxiety, highest-willingness-to-pay subject for Indian parents.

The five-year endgame

The long arc is not a subscription business — it is a **trusted mastery credential**. Assessment, backed by free-reasoning evaluation, that parents, then colleges, then eventually employers trust (the pattern of the Duolingo English Test, or a credit score). The moment our assessment becomes currency, we own rails others must run on. We plant this flag now even though it activates later, because it shapes every assessment decision we make today — every independently-mastered topic is a credential brick.

> The product's deepest job is not to deliver answers. It is to make a learner a better, more independent judge of their own understanding — and to do that as a daily ritual they return to because they want to, not because they are made to.

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The non-negotiable principles

These are the laws that settle decisions the rest of the document does not. When something is unclear, these win. They are deliberately few and deliberately strict.

Production-ready from line one

No MVP, no phases of the product, no throwaway versions. Built for scale and correctness from the first commit. We sequence features onto a complete foundation; we never ship a stripped foundation. We do not launch broken — we are not done until it is done.

The foundation is built complete even with no consumer yet

Where a platform service has no caller yet, its contracts, event schema, and ontology are still built complete and immutable from line one. Only the heavy service materialises when a consumer connects. This is discipline, not phasing.

Engineer compulsion only toward independence

The product is habit-forming — but it only ever engineers habit toward the behaviours that make a learner more independent: mastery, retrieval, independent attempts, consistency. The day a learner needs us less is the day we succeeded. Habit and integrity are the same loop, never opposing forces.

Spike to onboard, never to retain

Slot-machine mechanics are used only to onboard — to capture attention and deliver a fast first win. Retention is earned by a genuine ritual that makes the learner more capable. This single line is what separates us from Byju's.

Correctness is existential

No generated content reaches a learner unverified. A wrong answer shown to a child is not a bug; it is an existential failure. Simulators and mini-games are parametric, pre-verified templates — never freeform generation.

Compliant by design

Bulletproof from every direction — legal and data protection (including children's data), architectural, security, operational, ethical, reputational. Age and consent are first-class primitives that open legal capability; we build the doors the law provides and never chase gaps.

Felt, not claimed

Personalisation, care, and quality are demonstrated in the experience, never asserted in a paragraph. Felt personalisation beats claimed personalisation every time, and it is the cheapest deep trust we will ever buy.

Calm is the strategy

The whole category is red-urgency noise. We are the opposite — spacious, certain, quiet — and the calm is itself a status signal and the antidote an anxious learner and an exhausted parent are starving for. Restraint is the positioning made visual.

Sell the future, not fear

Roughly 80% aspiration, 20% honest concern. We show the next self — visible and close — far more than we warn of what is lost. For the competitive-exam segment the 20% is heavier, but the ratio holds.

One screen, one intention, one next action

Navigation is journey-based, not module-based. No learner faces a wall of menus; each surface answers what to do now, why, how long, and what it builds. Power is available, never imposed.

The most important of these is the relationship between **habit and independence**. The conscience layer of this product describes a calm, dependency-reducing academic companion; the growth engine is habit-forming and aggressive. These are reconciled by exactly one rule — engineer compulsion only toward the behaviours that make the learner independent — and they are never reconciled by abandoning either side. This tension is named again, in full, in §23, because it is the ethical spine of the whole build.

PART II

The Product

What the learner actually touches. Two entrances into one machine, an AI that composes the lesson per person, a realistic tutor, a separate engine that proves mastery, and an onboarding that feels like something is being cooked just for them.

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§ 03

Two doors, one machine

The product has two entrances, and they are not two features — they are two doors into the same machine. Both compile down to the identical structure: a personalised sequence of ontology nodes, generated and verified. This is the single most important architectural truth of the product surface.

The template door — acquisition and trust

On signup the learner enters age, academic board, and grade. We then show their board's syllabus by subject, by default. A grade-10 learner who picks Mathematics sees every chapter; each **chapter behaves as a course**; each course holds its topics; each topic holds the learning content and, towards the end, practice. This door is recognisable, board-aligned, "do my school" — and that is exactly why it is the **acquisition and parent-purchase engine**. It is the trust magnet: real, familiar school help.

The create-anything door — retention and moat

The learner can also create their own course on anything — hydro energy, rockets, the stock market, the physics of a game they love. The system keeps the learner's age and capacity in mind and builds content pitched right for them, with a clear path to go deeper if they want. This door is curiosity-driven, it is the thing the learner wants, and it is the thing Khan structurally cannot copy —

because it requires generation plus ontology plus verification working together. This is the **defensible half**. It is never a toy tab.

FIGURE 1 — BOTH DOORS COMPILE TO THE SAME THING

TEMPLATE DOOR

Board → grade → subject → chapter-as-course → topics

a pre-seeded path from the board ontology

CREATE-ANYTHING DOOR

Learner-defined course on any subject

a generated path mapped onto or extending the same graph

↓ **both become** ↓

ONE MACHINE

A personalised, verified sequence of ontology nodes the orchestrator composes per learner

The structure, precisely

- **Subject** = the container (maps to board → grade → subject in the ontology). It also acts as a folder.
- **Chapter** = the course unit (maps to chapter → topics, with sections plus practice).
- **Topic / subtopic** = the grain mastery and attempts are keyed to.
- **Practice** = a section at the end of every chapter, and a standalone capability per subject (see §06).

The fetch, and the fallback

Roughly 95% of the time we fetch the board and its latest syllabus automatically — this is the ontology ingestion pipeline doing its job. For the 5% where we cannot, the learner creates a **Subject** (which acts as a folder) and **Chapters** (which act as courses, with all the sections to master that chapter plus a practice section at the end). Note what this means architecturally: the fallback is the create-anything path. A user-created subject is just an un-seeded ontology subtree that the content engine populates on demand. There is no separate code path — one machine, two entrances, and the fallback is the second entrance wearing a school uniform.

> Design each door for its actual commercial job: the template door is built to **acquire and earn the parent's trust**; the create-anything door is built to **retain and defend**. They are one machine underneath and two products on the surface, and we never let the second one degrade into a novelty.

§ 04

The learning experience & the orchestrator

There is no fixed order of content. The motive is a genuinely personalised learning experience, and the way we deliver it is an AI orchestrator that decides — per learner, per topic — how to convey it: which modalities, in what sequence, with how many repetitions, until the learner has understood. The order is not a rail. It is an output.

The full palette of modalities

We do not provide one type of content for everyone. For any topic, subtopic, or chapter the orchestrator can draw on readings, videos, quizzes, simulators, flashcards, worksheets, interactive content, podcasts, mini-games, and more — in whatever blend and order makes that specific learner understand. It is a blend of all, focused differently depending on the learner.

> POSITIONING LANDMINE — HANDLE PRECISELY
>
> We do **not** market on "learning styles" (the visual / auditory / reader split, VAK/VARK). That theory is thoroughly discredited — matching instruction to a supposed style does not improve outcomes — and for a company staking its whole positioning on results and a trusted credential, marketing on a debunked theory is a clean shot any informed parent or competitor can take at us. The instinct underneath it is correct, but it is mis-named.

What we do instead is **adaptive multi-representation**, which the evidence actually supports (dual coding, the multimedia principle): every learner gets the concept in multiple forms, because richer encoding builds more retrieval routes — and that helps everyone, not because people are locked to one channel. The blend and emphasis adapt to three real signals: the nature of the concept (geometry is spatial, long division is procedural, a definition is verbal), the learner's current mastery and which gap is firing, and the learner's engagement preference as a motivation lever. Preference is legitimate — a learner who enjoys video spends more time, and time-on-task is real. The false claim we avoid is that style-matching improves comprehension. We keep the whole rich palette; we change only the justification, and that protects the credential thesis.

Active-first — the rule that protects the difference

A wide blend of readings, videos, and podcasts tempts the system back toward consumption — watch, read, listen — which is precisely the passive, explain-first pattern that would make us a Khan clone with a chat button, and Khan is free and excellent. So a hard rule governs the orchestrator: **the opener is always active.** A topic opens on a problem, a prediction, or a challenge. The learner attempts and struggles. The passive modalities — video, reading, podcast — are the reveals and scaffolds that fade, summoned after productive struggle has done its work, never served as the opening act. Quiz, worksheet, and simulator-as-challenge are struggle; video and reading are reveal. The universal loop is pose → struggle → reveal, and the inverted order is the entire reason someone pays.

Video is not the spine

Per-learner bespoke video is slow, expensive, passive, and competes head-on where Khan already wins. Our primary explanatory modality is **Vidya annotating a live interactive canvas** (§05) — strictly better than video because it is interactive and responds to the learner. Video, where it appears, is short and generated: nuggets of typically **15–30 seconds** for the best experience, with **60–120 seconds** reserved for genuinely irreducible single concepts. No fixed number; it varies with the requirement. Short nuggets are not only better pedagogy — they are cache-friendly and shareable, which makes them cheap to serve and useful as marketing raw material.

What the orchestrator actually decides — and why it is the moat

On each step the orchestrator reads the learner's mastery dimensions on the current node, which of the ten gap types is firing, the nature of the concept, the learner's preference and spaced-retrieval state, and outputs the next modality, its difficulty, and whether to repeat or advance. Early on this is a well-designed **heuristic policy** over those signals — it does not pretend to be a learned model on day one. Over time it becomes a trained model (the platform's Track 2), because every modality served and the mastery delta it produced is a training row in the event lake. That is the compounding moat: the policy gets smarter retroactively across every learner.

Three reactive rules, proven by the field

Three behaviours the strongest tutors in the world have already validated are first-class in our policy (see the competitive read, §27):

- **Switch the representation on failure.** When a learner stumbles, the move is not "try again" or only "drop the difficulty" — it is to change the representation and come at the concept from a new angle, live (a fraction shown as pizza slices becomes a number line). Adaptive multi-representation is not only an opening blend; it is a reaction to being stuck.
- **Select forward, not just next.** The orchestrator picks the problem that best prepares the learner for a future harder problem — not merely the next node in sequence or the most recent gap. It optimises the path, not just patches the hole.
- **Detect frustration before disengagement.** Guiding by question is the default, but dogmatic Socratic questioning makes learners quit and retreat to passive video. So the policy watches for frustration and switches to reveal or work-with-me before the learner checks out. Pose → struggle → reveal: we never withhold forever.

LOCKED — TERMINATION CONDITION

"Until they understand" must be bounded or it is infinitely expensive. **Understanding is defined by the mastery band crossing independent on unaided evidence from Practice.** The loop serves modalities until the evidence says stop — not until the AI feels satisfied.

Correctness and cost, built into the palette

Two modalities can never be freeform-generated: **simulators and mini-games must be a curated library of parametric, pre-verified interactive templates** the orchestrator selects and fills — never freeform model output, or we ship a broken or wrong simulation to a child. And the rich palette is only affordable because the artifacts are mostly assembled, not bespoke: modality cores live in a warm cache (verified once), cheap personalisation swaps the example and difficulty, and true bespoke generation is reserved for create-anything and genuine edge moments. This is the three-tier content economy of §19, and it is what makes "feels generated for you while reusing most content" both true and survivable.

Two superpowers, made first-class

The living plan. Khan gives a fixed tree; ours re-balances daily on real evidence and shows it — "you struggled with X, so I moved Y earlier." Personalisation the learner can watch happen beats personalisation asserted. **Interest-threading.** The interest profile is first-class and threaded through the examples — fractions via cricket scores, ratios via the game they play. This is hyperlocalisation at the individual grain, it is the curiosity drive doing the wanting, and it is structurally impossible for static content. Both adaptations are surfaced in tasteful one-liners, because felt beats claimed.

§ 05

Vidya — the tutor

Vidya is the signature interaction — a realistic on-screen tutor that guides, annotates, and behaves like a real teacher sitting beside the learner. This is where the "Brilliant, but more" vision actually lives, and where Khan has nothing. We concentrate our "another league" quality here, not in breadth.

What "realistic tutor" means — five things wired into one surface

- **A shared canvas** both the learner and Vidya can draw and write on.
- **Voice in and out** — the learner can talk to her and she responds.
- **She sees the learner's work** — the learner shows their working to the camera or canvas, and Vidya reads it. This is the preventive evaluation mode: correct it before you submit.
- **Graduated hints up the assistance ladder** — she gives the smallest useful nudge, never the answer, and the support visibly fades as competence grows.
- **Persistent memory** — she remembers what the learner struggled with last week, picks up where they left off, references their wins. This is the thing Brilliant does not have, and for a young learner a tutor that knows them is emotionally sticky in a way utility never is.

That fifth property is also quiet switching cost. Every session deepens Vidya's model of the learner; the longer they use her, the more they would lose by leaving. This is the Hook-model investment phase, accumulating invisibly (see §13).

The architecture — what "reacts to natural language, understanding context" actually requires

Vidya is not a chatbot with a face. She is a real-time multimodal tutoring loop, and that is frontier-hard. Five capabilities have to work together — and most of her intelligence lives in how well we assemble them, not in the raw model.

1 · Perceive

WHAT She reads the learner's work — canvas strokes, a photographed page of working, the on-screen state — hears their voice, and reads their text, fused into one picture of what this learner is doing and where they are stuck.

WHY This is the keystone and the riskiest part. If she cannot reliably read messy handwritten math and understand Hinglish and accented speech, nothing downstream works. It is the same capability the grading-calibration harness gates — which is why the first build action is a perception spike, not a chat demo.

2 · Understand context — four layers

WHAT A context assembler pulls all four layers into every call: the **turn** (what they just said or did), the **session** (the last several steps), the **lifetime** (they struggled with this exact thing last week; they love cricket), and the **curriculum** (the verified correct method for this node).

WHY When a learner says a tutor "gets them," they mean this assembler is good. Her intelligence is mostly the quality of the context we hand the model — drawn from the session buffer, the KGtoPG mastery and gap state, persistent memory, and the verified solution.

3 · Reason pedagogically — grounded in correctness

WHAT She picks the move — Socratic question, smallest hint, switch representation, reveal, encourage, back off, advance — as a live policy up the assistance ladder, never a script. And she reasons against the verified solution, never free-styling the mathematics.

WHY A tutor that confidently teaches something wrong is the one existential failure (§02, §19). Vidya is wired to the verification substrate and always reasons against a known-correct solution. The representation-switch-on-failure and the frustration guardrail (§04) both live here.

4 · Respond naturally

WHAT Voice and annotation that feel present: sub-second to first response, interruptible so the learner can barge in, context-referential ("you flipped the sign here"), drawing on the canvas in sync with speaking.

WHY Turn-taking and interruptibility are most of what separates "alive" from "walkie-talkie." Latency over a second reads as dead.

> ### 5 · Remember
>
> WHAT She writes back every session — what was attempted, the misconception, what worked — so she resumes where they left off and the relationship compounds.
>
> WHY That continuity is the emotional stickiness and the switching-cost moat — the thing no incumbent has.

The binding constraint — latency versus quality

Real-time multimodal voice tutoring sits at the edge of what is currently possible, so the loop cannot all run on a frontier model. Most turns — intent classification, hint selection, the high-frequency back-and-forth — run on fast, small models at the edge; only genuinely hard moments, like diagnosing a novel misconception, reach a frontier model. That routing is the two-track model gateway (§20, §22) doing real work, and it is also what keeps per-learner cost survivable — which is why Vidya's deepest mode reasonably sits behind premium.

> Build **this** transcendent before breadth. The atom of the whole product — the one topic that must be magic before anything scales — is, at its heart, Vidya teaching one concept beautifully: opened on a problem, build-don't-watch, graduated hints, live annotation, persistent memory, and clean evidence flowing to Practice.

> THREE FLAGS THAT GOVERN HOW VIDYA IS BUILT
>
> **Perception is make-or-break, and is proven first — narrow.** One subject, one device class, before anything else; every other Vidya capability rides on it. **Correctness grounding is non-negotiable** — she always reasons against a verified solution, never free-styles. **Vidya is the one exception to "fully functional everywhere on day one."** Everywhere else we build complete from line one; Vidya specifically must be transcendent on the atom — one math topic — before she goes wide, because she is the hardest and most expensive surface. The atom essentially is Vidya teaching one concept perfectly: get it right and breadth is engineering; get it wrong and breadth is just scaling a disappointment.

> THE GATE BENEATH VIDYA
>
> Vidya seeing a learner's work and giving graduated hints is **free-reasoning evaluation wearing a friendly face**. If that grading is not reliable on messy young-learner reasoning (Indian-English, Hinglish, vernacular, handwritten working), the signature interaction is not reliable. The grading-calibration harness (§19) is the keystone, and the eval-harness spike is the first thing we prove when we build (§25).

§ 06

Practice — the evidence engine

Practice is not a quiz tab. It is the evidence engine, and it is the addition that makes everything else work. Learn proposes; Practice proves.

Practice is where independent mastery gets demonstrated (the keystone dimension), where the learning-gap engine fires, where spaced retrieval against the real forgetting curve lives, and where misconception detonation lands. It is the loop that feeds the knowledge twin, re-balances the plan, generates the parent artifact, and lays the credential bricks. To the learner it is framed plainly: **this is where the system finds out what you actually own, alone.**

The two shapes of Practice

- **End-of-chapter practice** — a section at the close of every chapter, generated at the right difficulty, targeting the learner's actual misconceptions, not pulled from a static bank.
- **The standalone practice capability, per subject** — the learner picks as many chapters or topics as they want via a multi-select, and the AI curates a custom practice course or session to master exactly that combination. This is the practice orchestrator drawing on mastery, the ten gap types, and the spaced-retrieval schedule across the selected nodes.

How it behaves

Practice formats stay varied to keep it alive. Spaced repetition schedules review when memory is genuinely fading — accurate, never guilt-based. Every attempt contributes evidence to the mastery profile, not just a completion tick — which is why the band can only cross to independent on unaided Practice evidence. When the gap engine identifies the specific broken mental model behind a wrong answer, Practice is where **misconception detonation** lands: the one counterexample engineered to break that exact misconception, remediation that is aimed rather than generic.

§ 07

Mastery, gaps & evidence

The platform never reduces a learner to an average score. Two 80%s are not equal — a score earned with help, on easy questions, long ago, is not the same as one earned alone, on hard questions, recently, and repeated. This model is inherited from the KGtoPG contract and is the same one across the ecosystem.

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Mastery = Performance × Reliability × Independence
× Difficulty × Recency × Consistency
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The model is a **product, not a sum** — a near-zero on any dimension pulls the whole reading down. That is the design intent: being unable to work independently caps mastery no matter how strong the other dimensions are. **Independence** is the keystone almost no system tracks — it separates what a learner can do alone from what they can only do with help. A learner at 90% with heavy hints and 62% unaided is not "developing at 76%"; they are support-dependent, and the platform says so. Learners never see the raw formula — they see plain-language bands: not-started, emerging, developing, secure, independent.

The ten gap types — naming what is actually wrong

A low score is a symptom, not a diagnosis. The engine distinguishes ten kinds of gap, because each needs a different response, and the gap that is firing is one of the inputs the orchestrator reads when it chooses the next modality.

- **Prerequisite** — an earlier concept is weak
- **Conceptual** — the underlying idea isn't understood
- **Procedural** — the idea is clear but the steps fail
- **Application** — fine on familiar questions, lost on new ones
- **Retention** — understood once, now forgotten
- **Language** — the concept is fine; the wording is the barrier
- **Accuracy** — method is right, careless errors recur
- **Speed** — correct, but not within the time
- **Confidence** — knows it, avoids or second-guesses
- **Support-dependency** — strong with help, weak alone

Evidence discipline

A gap is never confirmed from a single bad score, and no permanent judgement is made from one interaction. Every conclusion about a learner is linked to the evidence behind it (the Evidence Graph), and the profile updates only when fresh evidence is collected. The knowledge profile is queryable by the learner — what am I weakest at, what should I do next, what unlocks this harder thing — and it shows independent versus support-dependent mastery, not just a number. This is the metacognition thread: the platform's deepest job is to make the learner a better judge of their own understanding.

Performance graphs, predicted scores, and trajectories

For each chapter and subject we show performance graphs and charts, a predicted score, and a trajectory — where the learner has been, is, and is heading, plus where they could be if they continue at this pace. We provide tailored, curated learning plans to close the gap to their goal.

Efficacy instrumentation — from line one

Because mastery is measured on real, unaided evidence, we can measure something most of the field cannot: the **pre-to-post mastery delta** a learner actually gains. We instrument that from the first commit. In a market where parents have moved from fear-of-missing-out to fear-of-being-scammed (§18, §27), a real efficacy number — "learners who finish this chapter improve from emerging to secure on average" — is the strongest, most defensible thing we can show, and the foundation of the credential thesis. The discipline is the same as everywhere: framed as honest ranges grounded in evidence, never as a guarantee, never a manufactured number.

> TWO FLAGS ON THE ANALYTICS
>
> **The trajectory chart is a sales surface, not a stat.** "Where you'll be at this pace vs where you could be" is future-pacing — the capability-gap conversion lever rendered as a chart. The gap between the two lines is the upgrade pitch; design it as a conversion moment shown at a milestone (see §16), not a passive widget. And **predicted scores are a reputational liability if stated as numbers.** Predict 92%, deliver 70%, and trust breaks with the exact parent who screenshots everything. Frame as honest ranges with visible uncertainty, and potential as conditional ("if you close these three gaps") — never a guarantee.

§ 08

Custom courses & competitive exams

Learners can create as many courses as they want on anything outside their syllabus. But creating is not the point — creating, then completing, then progressing is. The product is opinionated about that.

Custom courses — built with a completion bias

> ### Age- and capacity-aware generation
>
> WHAT When a learner wants to learn rockets or the stock market, the system builds content pitched for their age and capacity, with a clear path to go more advanced or in depth if they choose.
>
> WHY A twelve-year-old and a sixteen-year-old asking for the same topic need different pitches; offering all of it without judging the level produces content that is useless or overwhelming.

> ### Scoped into a finishable arc
>
> WHAT A created course is not an open-ended void — the orchestrator scopes it into a real course with a visible end and the same boss-battle capstone as a template chapter.
>
> WHY A created course that is abandoned is worse than one never created — it teaches the learner the product does not finish what it starts. The completion bias is built in: the system recommends the right depth, flags when a learner is over-reaching their prerequisites and bridges them there, and nudges them back to in-progress creates instead of letting unfinished courses pile up.

> Creating is the hook; **completing-and-progressing is the value.** We offer all of it, and we are smart enough to recommend — because the goal is not a library of half-built courses, it is mastery the learner actually reaches.

Competitive exams — as Subjects

Any competitive exam is available as a **Subject** in its own right, with the relevant syllabus mapped in — an exam is just a goal-oriented ontology subtree. Because the platform is smart, it guides: "you have mastered this chapter or topic — you can skip, revise, or redo." The end goal is explicit: **ace the exam.** The readiness forecaster projects the likely outcome and recalculates what is achievable given the time left and each topic's weight; the revision planner rebalances on real progress rather than nagging that tasks are overdue. We help the learner reach their goal, and we steer them past what they already own.

BUYER-PSYCHOLOGY NOTE

The competitive-exam segment (JEE/NEET foundation, NTSE, Olympiads at this age) is the **highest willingness-to-pay and the most fear-soaked** corner of Indian edtech — the hardest place to hold "calm, no-fear." We win there by being the calm, capable alternative that still delivers outcomes; the aspiration framing has to gesture harder at results here, and the 20% honest-concern share runs heavier. How prominent competitive-exam is at launch versus after the core proves is an open sequencing decision (§26).

The end-of-chapter report — one artifact, three jobs

At the end of every chapter the learner gets a report that shows everything they learnt, how much they mastered, and all the metrics. The same generated artifact does triple duty: it is the learner's **record** of mastery, the shareable **proof-loop media** (§18), and the **trophy** minted into their trophy room (§14). Build it once, beautifully.

§ 09

Onboarding — the cooking experience

Onboarding is the highest-leverage surface in the product. For a viral free product, the first sixty seconds decide everything — most signups bounce before they learn anything. So onboarding is the first aha, not a form. It should feel like a master chef tasting and adjusting, not a registration desk — a sense that we are cooking something amazing and tailored to them.

How it behaves

> ### Fixed at the start, adaptive within a few answers
>
> WHAT It opens as a fixed flow but reshapes the next question based on each response, so by the third question it already feels personalised.
>
> WHY That is the personalisation promise delivered before the learner has even started learning — felt, not claimed.

> ### Indirect data-gathering, disguised as curiosity
>
> WHAT We ask innovative, interesting questions that get to know the learner without explicitly asking what we want to know. We don't ask "what's your level" — we pose a delightful problem and read how they approach it. We don't ask "are you exam-anxious" — we ask something playful about how they feel before a test.
>
> HOW Some questions exist purely to make the learner feel seen and cared for — "if you could instantly understand one thing, what would it be?" Even where we do not use the answer mechanically, it is still knowledge about that learner, and more importantly it makes the product feel like it cares, which is the cheapest deep trust we will buy.

> ### Ends on the map, already partly lit
>
> WHAT The flow ends on a fast, delightful diagnostic that opens the learner's map already partly lit — "here's everything you already know, and here's your path."
>
> WHY This is endowed progress — the onboarding aha — and it seeds the knowledge twin and the archetype read from minute one.

For minors, the parent's consent step (handled over WhatsApp, §17) is woven in without breaking the learner's flow — the child keeps cooking while the parent grants consent in parallel.

PART III

The Psychology & Growth Engine

The part that makes it spread and stick. The free model designed properly, the mechanics embedded as product, the layer the user never sees, the social and status system, the generosity engine, the conversion architecture, the parent engine, and how we enter the market.

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§ 10

The free model & the session

The product is free, like Brilliant — the learner gets a limited amount per day, and then is invited to go premium. But it is not a clock counting down and not a count of sections. It is a behavioural system that reads the learner and ends the free day at the right emotional moment. This is the spine of the whole monetisation model, so it is designed in full here.

It is not a limit. It is a session — a workout.

A gym-goer does not train for six hours; they do a strong session and come back tomorrow, and that is the optimal training pattern. So the free daily cutoff is the app being a good coach — "that was a strong session, rest and let it sink in, here's what's waiting tomorrow." This reframe is not spin; it is the actual science (spaced practice beats cramming, ending on a win consolidates, rest is when learning sets). And it collapses four things we want into one screen: the freemium cutoff, the anti-streak / celebrate-rest mechanic, "optimise for outcomes not screen time," and the cliffhanger. The free limit is sold as pedagogy because it genuinely is pedagogy — which is the structural opposite of coercion, and the answer to "why doesn't this feel like a paywall."

The meter is invisible and behavioural

"Some AI that understands user behaviour" means behavioural telemetry — not a clock, and never reading a learner's face. The engine reads, per session: mastery events landed today (has the aha happened yet), flow signals (latency falling, accuracy and fluency rising, hint usage dropping), the early fatigue signal (latency creeping up, accuracy dipping, longer pauses), the anticipation state of the next node, the lifecycle stage (day-N in their journey), and archetype. From those it computes one private **momentum reading** the learner never sees.

The cutoff is timed to the crest, not the fatigue

The precise moment matters. We do not cut "just before the peak" — cutting before a win leaves the learner unsatisfied and the memory curdles into "not worth it." We cut **on a liking peak while wanting is still high**: right after a genuine comprehension win, in flow, before fatigue, with the next reveal visible but unopened. By the peak-end rule, the ending defines the memory of the whole session, so we engineer the ending to be a high with an open loop — and that open loop is the cliffhanger that pulls them back tomorrow.

> LOCKED — THE CUTOFF POLICY & THE ETHICAL GUARANTEE
>
> Fire the cutoff when at least one real mastery win has landed today, momentum is at or near crest, and an unopened high-anticipation node sits ahead — **never mid-struggle, never before the first win, never as punishment.** The hard guarantee, coded into the policy: the cutoff can only ever fire after the learner has had a real win that day. A free learner always learns something real, every single day, before anything is gated. A free tier that frustrates cannot go viral; this guarantee is what keeps free genuinely good.

Two peaks, never conflated

The **session peak** ends every day on a high — soft, generous, always. The **lifecycle peak** is when the hard premium ask lands hardest — and that is not day one. For the first roughly two weeks the cutoff stays loose and the ask is nearly absent: we let identity and ritual form, maximise the wanting phase, and buy word of mouth. The hard conversion ask sharpens from week three, when the habit is built, the twin shows real progress the learner would lose, and loss aversion is honest rather than manufactured. The budget is a **lifecycle dial** —

loose when we need word of mouth, tightening as the base grows and the learner is genuinely hooked.

> The exact threshold on that dial, at each lifecycle stage, is **the single most important number in the company.** We tune it on cohort data; we never guess it.

The cutoff screen is the most commercially important UI in the app

This one screen is where retention and revenue are both won, so it gets more design love than anything else. It is never a wall and never a countdown. It is a designed peak-end moment: a calm, premium celebration of exactly what the learner just mastered (the knowledge twin igniting the region they just lit), a one-line cliffhanger naming tomorrow's unlock by its most intriguing face, and — at the right lifecycle stage — a soft door to premium framed as "train as long as you want." The feeling to leave them with is **pride plus anticipation**, never restriction. It deserves a dedicated prototype before almost anything else.

§ 11

The psychology engine

Every mechanic below is embedded as product, not bolted on as a campaign. Each is listed as the principle and how it lives in the experience. The arc we are running is Byju's ignition fused to Brilliant's retention — manufacture demand and convert at the peak, then hold with a genuine ritual — with one hard line: spike to onboard, never to retain.

The retention and habit mechanics

- **Cliffhanger cap** (Zeigarnik + peak-end + variable anticipation + implementation-intention) — stop one step before the reveal; the cap is a save point and the conversion moment, not a wall.

- **Endowed progress** — credit what the learner already knows via the diagnostic; the map opens already lit.
- **Goal-gradient** — many short arcs that are already ~80% done; effort accelerates near a goal.
- **Identity over motivation** — convert fleeting motivation into durable identity in the first ~2 weeks: "learners," identity streaks ("24 days of being a learner"), milestones as titles of self.
- **Build-don't-watch** (generation effect + productive failure) — pose → struggle → reveal; predict-then-check, drag-to-assemble proofs, fill-the-missing-step, teach-it-back to Vidya (protégé effect); scaffolding visibly fades.
- **Hook model** — every attempt trains the learner's model and improves their experience; use-improves-product is switching cost.
- **Honest loss aversion** — the real forgetting curve via spaced repetition, never Duolingo guilt.
- **Ethically-bounded variable reward** — variety in content and celebration, never loot boxes; variable reward attaches only to genuine achievement.
- **Aesthetics as psychology** — processing fluency: a premium feel makes learning feel achievable. Load-bearing, not decoration.

Wanting vs. liking — the neural crossover

Desire and pleasure are different systems. Weeks 1–2 maximise **anticipation** (wanting); from week 3 every session must deliver real **consummatory satisfaction** (liking) or wanting curdles into anxiety. The ratio of anticipation-moments to satisfaction-moments should invert across the first month — this is the slot-machine-to-ritual crossover at the level of the brain, and it is why the lifecycle dial in §10 tightens when it does.

The two share loops

The challenge loop — acquisition

Share the hard problem, never the answer; WhatsApp-native, attemptable in-thread. A learner who just cracked something wants to make a friend struggle with it. Near-zero CAC.

The proof loop — credibility

The beautiful branded mastery artifact (the chapter report / trophy). Child-triggered, parent- and peer-received. This is the credibility engine.

Referrals reward in learning, never cash. And the nuggets are the fuel for both loops — every sub-30-second mastery clip is shareable from birth, so modality production is also the near-zero-CAC engine's raw material.

Sell the future, not fear

Roughly 80% aspiration. Three honest gaps, all closeable by the product: **capability** (the version of you that mastered it — the knowledge twin, future-pacing), **status** (the founding-cohort and level system), **peer** ("someone just ahead of you" — a pull, never a demoralising shove). Always keep the next self visible and close.

§ 12

Crown-jewel mechanics

These are first-in-category and defensible — only possible with our engine, our mission, and our nerve. They are the mechanics nobody in edtech has shipped, and they are why the north star is reachable.

Misconception detonation

Name the broken mental model behind a wrong answer, then generate the one counterexample engineered to break it. Remediation that is aimed, not generic.

The conversational knowledge twin

Query your own cognition — "what am I weakest at," "what unlocks astrophysics" — and see not just lit vs. dark but independent vs. support-dependent mastery. The signature icon; generative hero art that breathes and ignites a region the moment something is mastered.

"Written for you" live problems

Personalisation as scarcity — a problem assembled for this learner, just now. Structurally impossible for static content.

Teach-to-unlock & the protégé economy

Pass a gate by teaching Vidya; great explanations are curated into content others see. The protégé effect made into a UGC economy.

Synthesis boss battles

A live problem assembled from everything the learner knows — the chapter capstone. Pedagogically a consolidation; experientially a triumph; an output it mints a trophy. Not a free-month dispenser (see §15).

Sanctioned rabbit holes

The content engine bridges from where the learner stands to any concept they are curious about — curiosity rewarded, never blocked.

Anti-streak

Celebrate planned rest. The contrarian, ownable, scientifically honest line that separates us from guilt-driven streak mechanics.

Free-reasoning as primary input

The keystone that makes all of the above possible — the learner reasons in their own words and we evaluate it. The hardest thing to get right, and the moat if we do.

§ 13

The invisible layer

A great deal of the product is machinery the learner never sees and never understands — it should just feel simple and premium while the intelligence works hard underneath. The principle that keeps this powerful and keeps us off the predatory path comes from one distinction.

> Borrow Instagram's machinery; refuse Instagram's objective function. Instagram's engine optimises one thing — time-on-app. Ours optimises **mastery-then-leave.** Same invisible intelligence, opposite goal. "The user doesn't understand why this showed up" is wonderful when what shows up is the right next thing for their learning; it is predatory when it is the thing most likely to keep them scrolling. We do the first, always.

What we take from Instagram

- Per-user **optimal-timing notifications** — fire in this learner's historical active window, not a blast time.
- **Frictionless onboarding** to a fast first win.
- A **personalised next-best-action queue** — their version of the feed, but a queue of the right next thing, never infinite scroll.
- Tasteful **social proof** — "someone just ahead of you cracked this," a pull, never a shame.

What we refuse, on purpose

- **Infinite scroll and autoplay-next** — maximises time-on-app, the opposite of optimising for outcomes.
- **Variable reward on consumption** (the slot machine) — we attach variable reward only to genuine achievement.
- **Guilt / FOMO notifications** — and this one needs the honest answer below.
- **Engagement-over-outcome** as the objective function — the core Instagram goal is the wrong goal for us.

THE HONEST VERDICT ON GUILT / FOMO

They work — and we still refuse them, because we have a stronger lever for the same job

Guilt and loss-framed notifications measurably lift short-term return; Duolingo's guilt-owl is famous because it works. But they work by manufacturing anxiety, they erode brand affection over time, and aimed at minors and stressed parents they are actively harmful and dead against our calm positioning — a reputation landmine. We don't need them, because we have a stronger lever: **anticipation and pride.** Pull the learner back with something they want to return to — the cliffhanger they were left on, the next reveal pre-loaded, the twin that grew, the streak-as-identity — not fear of what they'll lose. Same re-engagement outcome, opposite emotional mechanism, and it compounds brand love instead of spending it down. The only loss-aversion we use is the honest one: the real forgetting curve.

The invisible mechanics that earn their place

- **The behavioural peak-cut meter** (§10) — momentum read from latency, accuracy, hint-usage, fatigue onset, anticipation. Never shown.
- **The archetype engine** (§18) — six archetypes detected in week one, silently re-scored, driving notification copy, reward type, paywall framing, and timing per person. **Adapts framing, copy, and timing — never price.**
- **First-session engineering** — the first session is hand-built to guarantee an aha within minutes; this is where viral signups are won.
- **The diagnostic that opens the map partly-lit** — endowed progress; feels like recognition, functions as motivation.
- **Goal-gradient acceleration** — quietly raise reward density and visible progress as a chapter nears completion.
- **Pre-loaded cliffhangers** — the system always has the next high-pull node teed up and visible-but-locked at session end.
- **The investment moat** — every attempt tunes the learner's twin and tutor, so the longer they use it the more they would lose by leaving.

- **Processing-fluency priming** — the premium calm UI makes learning feel achievable. They just feel "this is nice and I can do this."

§ 14

Status, social & identity

Comparison without toxicity. The instinct is right — learners want to see where they stand and flex what they have achieved — but it must never become a usage-ranked leaderboard that shames the slow and rewards the merely busy. The whole system is built on one rule: status flows from learning, never from time spent.

XP and levels — the spine

XP accrues only from learning signals

WHAT Experience points come from independent mastery, boss-battles cleared, and streak consistency — and from daily streaks contributing to that XP. They never come from raw time or usage.

WHY Rewarding time is the Instagram objective-function trap; rewarding mastery keeps the status economy pointed at outcomes. This is the §02 principle made concrete in the points system.

Level is a status identity, not a rank

WHAT XP rolls into a Level — the public number a friend sees. It is not a leaderboard position. It does not have to be competitive on usage; a level is enough.

WHY A friend who is "Level 14" is aspirational, not a verdict on you. Level pulls without shaming — it is the status gap as a draw, not a humiliation.

The trophy room — the flex surface and the virality engine

Every boss-battle cleared mints a **trophy** — the chapter's shareable mastery report (the same artifact from §08). The trophy room is where all of a learner's achievements live and where they flex them. Finishing more boss levels earns badges and a boost in points; daily streaks feed XP too. Friends can see each other's trophy rooms, and trophies are screenshot-and-share-native, which feeds the proof loop directly. This is deliberately about what you mastered, not who you beat.

Friends — mastery-visibility, not a ladder

Learners add friends to see mastery progress, levels, points, trophies, and streaks. There is a gentle "someone just ahead of you" pull, opt-in in spirit. There is **no global leaderboard at launch** — global ranking breeds the exact anxiety we are positioned against, and it is where the Exam-Anxious archetype churns. If competition appears later it is opt-in cohorts, never a default ranking.

The batch badge — a quiet masterstroke

> **"Class of '26", member since 2026**
>
> WHAT The year shows in the learner's profile and as a badge on their avatar — a "26" mark — plus "member since 2026" in the profile.
>
> WHY It does three jobs at once: it manufactures founding-cohort status (the velvet-rope feeling without any gate), it creates belonging (the Belonger archetype's home), and it ages into loyalty — a "Class of '26" member three years in has identity sunk into the brand.
>
> HOW Make the founding years visibly special, and **never re-issue early badges** — scarcity is the whole point. The earliest cohorts must always be visibly the earliest.

§ 15

The grant engine & scholarships

No free trials, ever. We charge nothing until a learner subscribes to premium. Generosity is a weapon, and we use it aggressively — but never as an advertised, grindable reward. Free rewards are not fraud; they are hardcore, tactical marketing in a cut-throat market, and we deploy them precisely.

Discretionary comp — the Anthropic method

The free month is **not** a default the learner keeps trying for or even knows they can earn. There is no boss-battle-vends-a-free-month mechanic — a known, grindable reward turns a calm product into a carnival game and corrupts the very mastery signal we depend on. Instead, the system smartly presents premium to the right learners when it makes sense — silent, behaviour-triggered generosity, modelled on how Anthropic gives discretionary credits rather than a trial.

FIGURE 2 — HOW A GRANT HAPPENS

QUALIFYING SIGNALS	daily-streak consistency · full independent chapter mastery · boss-battles cleared · exam-season proximity · depth of engagement

↓

GRANT-WORTHINESS SCORE	a private score the learner never sees — computed continuously from the signals above

↓

THE SURPRISE	on crossing the threshold, the learner is gifted a few days or a month of premium — framed as recognition and care, never as a reward they chased

Three properties make this work. It is **invisible until granted** — no grinding, no expectation, no gaming. It is **targeted at our highest-intent learners** — dedicated learners are the best conversion and advocacy bets, so the comp is precision spend, not a blanket giveaway. And it runs on **taste-then-lose** — they live in premium for a while, the twin shows everything they built there, and when the grant lapses, honest loss aversion converts them. Dedicated learners and exam-time learners are exactly who we feel generous toward; the data points the app gathers tell us who and when.

LOCKED — THE BOSS BATTLE'S ROLE

The synthesis boss battle survives as a pedagogical capstone, a shareable report, and a trophy. It does **not** dispense free months. Generosity is decided by the discretionary engine on behaviour, not vended by a game the learner can farm.

Scholarships & Sponsored Seats — generosity made institutional

Beside the silent engine sits the overt, applied-for path. A learner applies to a scholarship program, and a **Patron** — a company or a CSR fund — grants their premium for up to 6–12 months. The learner sees who sponsored them ("your year of premium is funded by [Patron]"), and sponsors get optional branding if they want it.

- **Why it is strong** — it is genuinely fundable (Indian CSR education spend is large and hungry for clean impact stories), it converts our generosity into a B2B revenue line and a brand-trust signal at once, and it scales reach without discounting the premium perception.
- **The hard design rule** — a sponsored learner's experience is **identical** to a paid premium learner's. Never a lesser "charity tier," or it shames the learner and poisons the goodwill.

Why the integrity layer protects all of this

Being aggressive with generosity only works if the generosity reaches humans. The integrity / anti-abuse layer (§19) is not a brake on generosity — it is what protects its return: without it, bots, multi-account farmers, and Sybil rings siphon the free premium and referral rewards meant for real learners, and fake signups pollute the very behavioural data we steer growth by. Integrity is how hardcore generosity actually lands on real learners instead of leaking to abusers.

§ 16

The conversion architecture

Free by default; we cash on a single premium plan. The boundary between free and paid is dynamic, the plan is one clear identity upgrade, and every ask lands at an emotional peak — never on a timer, never mid-struggle.

The plan — a single premium identity

One premium plan. The working name is the **Superstar upgrade** — naming the tier as an identity the learner becomes ("become a Superstar"), not a feature bucket ("upgrade to Premium"). That principle — identity-naming over feature-tiering — is locked, because it is identity-over-motivation applied to monetisation, and it is warm and aspirational.

> **OPEN — PRESSURE-TEST THE EXACT WORD**
>
> "Superstar" is a strong candidate, but it reads slightly young for the 16–17 competitive-exam end of the range, and it is a common word that is hard to own distinctively. The principle is locked; the exact word stays open, to be set next to two or three alternatives that scale across an 11-to-17 span during the brand work (§25, §26).

The free / paid boundary — dynamic, generous, complete

Free is feature-rich and genuinely complete — the full template syllabus, the real Vidya, real Practice — metered each day by the behavioural session (§10), not crippled. A few deep capabilities are reserved for premium: the deepest Vidya (extended voice, live annotation, full persistent memory depth), create-anything beyond the template, the full trajectory analytics, and the richest parent reporting. The boundary is **dynamic** — it flexes by learner behaviour and lifecycle, using a modified version of the "give generously to grow" strategy. Free must always be good enough to go viral.

Pricing — anchored to a personal AI tutor, not to edtech

LOCKED WORKING FIGURES — TUNABLE ON COHORT DATA, BUT NO LONGER OPEN

₹999 / month · ₹8,999 / year (hero)

The annual is the hero, priced to feel like three months free (~₹750/mo effective), with the monthly sitting beside it as the deliberately-costly convenience option — that pairing is the decoy structure with a single plan. Annual-first banks cashflow, lifts LTV, and an annual buyer does not churn at the first wobble.

The anchor is the whole argument. We are not priced against Indian edtech (where Physics Wallah sits near ₹3,500/year); we are priced against a personal AI tutor. On that anchor, with general AI subscriptions averaging around $20/month, ₹999/month reads as a deal — and Brilliant itself sells globally at roughly ₹10,000–12,000/year and does fine. So ₹999/mo and ₹8,999/yr are defensible, arguably even conservative for what Vidya is meant to be.

The one real nuance is **who is doing the anchoring.** A self-driven, aspirational, older learner references AI tools — ₹999 is cheap. A parent paying for a younger child references tuition and Physics Wallah — there the number leans high, so for that segment the value has to be visible (the WhatsApp proof artifact, real efficacy), and the excellent free tier plus scholarships absorb the genuinely price-sensitive. The thing that makes or breaks the price is therefore not the number — it is whether Vidya feels like a ₹999 tutor at the moment of the ask. Which is exactly why the atom comes first.

The posture still holds: **do not buy reach by discounting the price** — that erodes the premium perception the calm-premium brand rests on. Buy reach with generosity — the excellent free tier, frequent discretionary grants, real scholarships. We tune the exact figures on cohort data, but they are the working numbers now, not placeholders.

The conversion trigger map

Every ask is fired at a peak, never by a clock, never mid-struggle. Hard asks are lifecycle-gated — nearly silent in weeks 1–2, sharpening from week 3.

#	TRIGGER	THE MOMENT / LEVER
1	**Session cutoff screen**	Daily peak-end; soft, lifecycle-gated; "train as long as you want."
2	**Cliffhanger**	At the unopened next reveal — premium opens it now.
3	**Discretionary grant**	The silent surprise (§15) — taste-then-lose does the converting.
4	**Create-anything attempt**	Feature gate at the point of intent — they already want it.
5	**Deep-Vidya reach**	When the learner reaches for the deepest tutor capability.
6	**Trajectory chart**	Current-pace-vs-potential at a milestone — the aspiration sell.
7	**Parent artifact + nudge**	Pride-then-opportunity over WhatsApp (§17).
8	**Taste-then-lose**	When a grant lapses — honest loss aversion.
9	**Post-aha high**	Right after a genuine win (variable, used sparingly).
10	**Identity milestone**	A level, a streak, a badge — "Superstars go further."

Payment surfaces: Razorpay on web, Apple and Google in-app purchase on mobile. **No trials and no charge until subscribe** — and therefore no surprise auto-charge dark pattern, which is also the honest, brand-protecting choice.

§ 17

The parent engine

The parent is the payer for younger learners and a powerful growth and retention force — but they are not the user, and they must never become a surveillance dashboard pointed at their child. The parent's product is a relationship, and it lives where parents already are: WhatsApp.

The parent's product is a WhatsApp conversation with a parent-companion AI

No install. Through one WhatsApp thread the parent gets the weekly visual progress artifact, milestone celebrations, an ask-anything channel ("how is my child doing," "what do I do at home," "explain this topic so I can help"), the pride-then-opportunity nudges, and the upgrade — paid in-thread or via a deep link. For India this is close to ideal: zero install friction, where parents already are, visual media lands, async and personal.

The notification pattern is concrete: tell the parent about achievements and progress, then — when it is true and earned — "they are at 80% of their capacity on this; here is how to push toward 95% via A, B, C using X and Y; the premium upgrade is how." Email is the backup channel for the same artifacts.

> LOCKED — THIS SURFACE ALSO SOLVES CONSENT
>
> The parent's WhatsApp onboarding is also the **verifiable-parental-consent surface**. That single design choice solves the consent architecture for minors — the parent "account" is a WhatsApp relationship (plus an optional light in-app view), and the consent it captures is what lawfully lights up the deeper intelligence for a minor (§19, §23).

> THREE HARD RULES ON THE PARENT ENGINE
>
> **Lead with pride, then opportunity — never deficit-shame.** Roughly 80% celebration, 20% aspiration framed as a strength to extend, never a failure to fix. The moment we shame a child to a parent to sell a plan, we are Byju's. **The "80% of capacity" number must be real**, computed from genuine evidence — manufactured to create anxiety, it becomes the manipulation we exist to replace. **Respect the channel** — WhatsApp Business has opt-in and template-compliance rules and polices promotional spam, so nudges run as compliant templates with consent, on a disciplined cadence (weekly plus milestones, never daily blasts). The best version of every nudge is child-triggered: "show mum what I just cracked."

§ 18

Go-to-market

We launch open, not invite-only — because we cannot tell a hot market to come back later, and we want to capture demand the moment it is warm. But "open" is engineered, not a free-for-all: it is wrapped in manufactured status and a viral queue so that openness still feels like a privilege.

Open access, engineered

- **Founding-member status skin** — scarcity-as-honor, not a gate. The batch badge (§14), the "Class of '26" identity, the visible specialness of being early.
- **Referral-position / waitlist-skip mechanic** — a viral queue where referring others moves you up; the queue is a growth loop, not a barrier.
- **Invisible wave-release** — onboarding waves tuned to the cost governor, smoothing load without ever telling a learner "no."

The reason invite-only is unnecessary for cost control: the **daily session meter is the cost governor** (§10). Because per-learner daily cost is already bounded by the behavioural budget, we do not need a gate to control spend — so we take the open, capture-the-market path and govern cost through the meter instead.

Trust as a feature — the post-Byju's wedge

The Indian market has moved from fear-of-missing-out to **fear-of-being-scammed** (§27): parents are now actively suspicious of edtech after the Byju's collapse. That makes trust a growth lever, not just an ethic, and we make ours legible in two ways.

- **A genuinely free, public, ungated content layer** that builds trust before any signup — the Physics Wallah move of earning belief with generosity first. A learner or parent can see the quality before they are ever asked for anything.
- **Visibly not-Byju's** — no sales calls, cancel anytime, no EMI loans, no surprise auto-charges, honest billing. The red flags parents have been taught to fear

become the things we conspicuously do not do, stated plainly where they can see them.

The archetype engine drives the go-to-market, silently

Six archetypes, detected in week one from behaviour and re-scored continuously. They adapt copy, reward type, paywall framing, and timing per learner — and **never price.**

Competitor Wants to win. Levels, boss battles, "someone just ahead."	**Mastery-Seeker** Wants to truly understand. Depth, the twin, independence.	**Exam-Anxious** Wants to feel safe. Calm, readiness, honest reassurance.
Ritualist Wants the habit. Streaks-as-identity, the daily session.	**Belonger** Wants to be part of it. Cohort, batch badge, friends.	**Dabbler** Curious, low-commitment. Create-anything, rabbit holes.

The growth loops, and the sequencing discipline

The near-zero-CAC **challenge loop** and the credibility **proof loop** (§11) run from launch, fuelled by shareable nuggets and trophies. The Scholarship/Sponsored-Seats program (§15) is a B2B2C and CSR cross-subsidy loop. Seasonal exam surges are demand spikes we ride. The data flywheel compounds underneath all of it.

> The discipline that protects all of it: **perfect the atom, then light the viral fuse — never viral-then-fix.** A viral loop on a broken core spreads disappointment at the speed of the loop. Depth-perfect-narrow before wide, always.

Build-in-public — deferred, with a clear angle

Build-in-public is a planned acquisition channel and gets its own dedicated working session — but the angle is set: a **faceless, artifact-driven** build-in-public where the story is the solo-founder-plus-AI build method and the thinking and decisions behind it, not feature demos or on-camera presence. Camera-shyness is not a blocker, and "building too fast to document" is backwards — the method is the rare, interesting story. Heavy guidance to come when we open that thread.

PART IV

The Foundation

What holds the weight. The subsystems that make the magic safe and affordable, the two-database architecture and the immutable event contract, the stack, the intelligence and admin brain, the compliance posture and the one tension that governs the ethics, and the conscience layer the whole thing answers to.

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§ 19

The first-class subsystems

Seven subsystems are accepted as first-class components, not afterthoughts. Each one is load-bearing — remove any and a headline promise of the product collapses. They are built complete from line one.

> ### 1 · Generation & cost economy
>
> WHAT A three-tier content economy: a warm cache of verified modality cores → cheap small-model personalisation that swaps example and difficulty → frontier bespoke generation reserved for create-anything and genuine edge moments.
>
> WHY It is what makes "feels generated and curated just for this learner, even though we reuse most of the material" both true and affordable. A viral free product cannot pay frontier-model prices per free learner; this subsystem holds gross-margin-per-free-DAU at a target and is the reason the rich palette and the giveaway strategy survive contact with scale.

2 · Consent- and age-tiered intelligence model

WHAT The deep engine lights up only at the consent and age tier the law permits; verifiable parental consent (captured over WhatsApp, §17) is what elevates a minor's tier. The un-elevated tier is constrained but still excellent.

WHY Children's-data law is the binding constraint on the whole intelligence layer; making consent and age first-class is how capability is unlocked lawfully rather than chased through gaps.

3 · Correctness / verification substrate

WHAT No unverified content ever reaches a learner. Symbolic computation (a CAS) checks mathematics; simulations are re-run to confirm; a second model cross-checks; a confidence gate refuses anything unverified and routes it for human or harder verification.

WHY Correctness is existential (§02). A wrong answer to a child is not a bug, it is a breach of the entire promise — and of the credential thesis.

4 · Child-safety subsystem

WHAT Moderation on any user-generated content (e.g. protégé explanations) before another learner sees it; crisis-detection on Vidya conversations; safety classifiers on dialogue; no private, unmonitored channels between users.

WHY We operate a social, generative product for minors. This is non-negotiable and reputational survival.

5 · Grading calibration harness — the keystone

WHAT The system that makes free-reasoning evaluation reliable on messy young-learner reasoning — text and voice, Indian-English, Hinglish, vernacular, handwritten working. Versioned rubrics, human-calibrated agreement, confidence-banded outputs, an "I think I'm right" re-grade path, adversarially hardened.

WHY Every mechanic in this document rides on it — peak-cut timing, the boss-battle gate, skip/revise guidance, predicted trajectories, the parent "80%→95%" number, the termination condition, the credential. It is the first thing proven when we build (§25). The framing is calibration — making the grading agree with human judgement — not doubt that it is possible.

6 · Prerequisite graph

WHAT An owned, expert-validated artifact — the directed graph of what must be understood before what. The content engine proposes edges; validation confirms them; a Khan-style skeleton seeds it.

WHY Prerequisite-gap remediation, sanctioned rabbit holes, and the living plan all depend on a prerequisite graph we trust. It is a moat precisely because it is validated, not merely generated.

7 · Integrity / anti-abuse layer

WHAT Sybil detection, referral-abuse and multi-account-farming defence, bot detection.

WHY It protects the ROI of an aggressively generous model (§15) and keeps the behavioural growth data clean. Without it, generosity leaks to abusers and the data we steer by is poisoned.

§ 20

Architecture & data

The architecture is ecosystem-grade from line one, even though Learner is the only consumer today. These invariants are inherited from the KGtoPG platform and hold across every product the ecosystem will ever ship.

Two physically separate databases

FIGURE 3 — THE TWO-DATABASE LAW

PLATFORM CORE · KGTOPG · FOUNDER-OPERATED

Canonical identity · immutable event store · ontology & prerequisite graph · profile · evidence & mastery · feature store · org/relationship graph

OPERATIONAL DB · PER APP

Each app's own operational data, isolated by multi-tenant row-level security

Cross-database foreign keys are forbidden · canonical references are opaque UUIDs validated through services · apps talk upward via a transactional outbox and read downward via governed views

The platform core is the canonical brain; each app gets its own operational database. They are never joined at the database level — a reference from an app to a canonical entity is an opaque UUID resolved through a service, never a foreign key across the boundary. This is what lets the ecosystem grow without entangling one product's data in another's, and it is why Learner can be built correctly today and inter-operate with future products tomorrow.

The event contract — non-negotiable from line one

Apps communicate upward through a **transactional outbox** (events emitted atomically with the writes that caused them) and read downward through **governed views** (never raw platform tables). The event envelope is attributed (app, canonical UUID, purpose, consent reference), append-only, schema-versioned, and free of raw PII. This is the highest-stakes seam in the whole system and there is no retrofit path — get it right on the first commit or pay forever. The event store is **immutable and replayable**: as the models improve, our understanding of past learners improves too, retroactively. The moat compounds backward, not just forward.

The two model tracks — never conflated

Track 1 is external market LLMs (Claude, Gemini, OpenAI), chosen per feature behind a model-agnostic gateway. **Track 2** is our own planned proprietary fine-tuned models and edge small language models — the margin and the moat — trained on the event lake. These two tracks are kept strictly separate in design and language; conflating "we call the best external model" with "we run our own models" muddies both the cost model and the moat story.

Conventions & deletion

- **API key convention** — `clss.<app>.<env>.<purpose>`
- **Deletion model** — on deletion, raw personal data leaves; de-identified and aggregated learning signal stays. Event-sourcing makes this clean and auditable.

The repo decision — what we lift from Classess School

The paused School codebase was audited. The verdict is **take / trim / torch**:

VERDICT	WHAT	WHY
Take	The contract layer — event envelope & primitives, the mastery contract, the ontology & prerequisite-edge artifact, the evaluation contract, the assistance ladder, consent primitives; the AI-fabric structure (router, verify, second-model, confidence-gate cache that fails closed, the Track-2 slot, capability registry, orchestrator); the DB migrations (events, PII vault, consent, governed views); the gateway wall (rate-limit → authn → RBAC → ABAC → consent → approval → child-safety → audit, deny-by-default).	Architecturally sound, hardest to rebuild, already the right shape. This is the **KGtoPG seed** for the clean Learner repo.
Trim	Institutional event families — attendance, timetable, substitution, parent-teacher meetings, teacher growth, institution policy.	Not needed for a learners-only product; cut to learning, mastery, evidence, content, consent.
Torch	All four role surfaces, the fourteen institution modules' product logic, the multi-tenant hierarchy UI, the old Classess 3.0 design system, ClassBoard, the React surfaces.	Institutional product surface is the wrong product; the Learner surface is built fresh on the v4 brand.

Net-new, not in the repo, built fresh: the FSRS scheduler, the daily-budget meter, the knowledge twin, the archetype engine, the crown-jewel mechanics, and the v4 brand.

§ 21

The technology stack

The stack is chosen, with the rule that the language is unified wherever contracts cross — TypeScript across contracts, gateway, modules, and surfaces, with a Python sidecar scoped to verification and the intelligence core.

Data & identity

Supabase — Postgres, Auth (phone-OTP first, plus Google/Apple), Realtime, pgvector, Storage. Redis for the budget meter, leaderboards, and session state.

Orchestration & models

FastAPI orchestration with a multi-model router (LiteLLM) — the Track-1 gateway. Free-reasoning evaluation and the misconception classifier sit here.

Content & knowledge

Plexus (13 engines) as the sole content source, RAG-grounded on NCERT and Indian-board catalogs (~3,500 items). The prerequisite graph as an owned validated artifact.

Mastery & memory

FSRS for spaced retrieval in the practice/evidence engine; Bayesian / IRT methods for the mastery estimate.

Surfaces

Expo (React Native), offline-first, plus a React web / PWA. The parent surface is WhatsApp (Cloud API) as a product surface, not just a notification channel.

Interactive & visual

Rive + Lottie + Framer Motion for motion; JSXGraph + Mafs for 2D math; Three.js / React-Three-Fiber for 3D. GeoGebra is eliminated. Remotion 4 for the video pipeline.

Payments & messaging

Razorpay (web) + Apple / Google IAP (mobile), Stripe where relevant. Resend (email) + WhatsApp Cloud API + push for lifecycle messaging.

Observability & ops

PostHog (analytics, flags, experiments), Langfuse (LLM observability), Sentry (errors), Infisical (secrets). Vercel + Railway/Render hosting, GitHub CI/CD.

Fonts & the offline path

Fonts are bundled locally from fontsource with no CDN dependency: Google Sans Flex, Fraunces, Inter, JetBrains Mono, Caveat. Offline-first reconciles with live generation by pre-syncing FSRS review packs and using edge small language models for offline evaluation, so a learner on a weak connection still gets a real session.

§ 22

AI, admin & the omniscient brain

AI-native is not a feature; it is the substrate. Content is generated, not stored; learners are understood, not merely recorded; experiences are adapted per person, not templated; and the system learns retroactively as its models improve.

The model gateway, again — because it is the margin

Everything routes through a model-agnostic gateway: Track 1 calls the best external model per feature today; Track 2 runs our own fine-tuned models and edge SLMs tomorrow — frontier models for genuinely hard generation, small models at the edge and offline for cost and latency. The split is the margin and the moat, and it is designed in from the start even where Track 2 is a slot waiting to be filled.

The super-admin cockpit — god mode, governed

A single control surface exposes every user and every metric, every marketing dial and every adaptive-engine dial, every cap threshold, every pricing lever, the content pipeline, every live experiment, and the data controls. It is governed by tiered access, an immutable audit trail, and break-glass procedures for the most sensitive actions — power that is total but never unlogged.

The omniscient brain

A natural-language-queryable, predictive understanding of any learner, cohort, or pattern — built on the event lake, the feature store, behavioural embeddings, and the knowledge graph, and governed by the same age and consent doors as everything else. Ask it who is about to churn, which cohort the new cutoff helped, which learners are grant-worthy this week — and it answers from the canonical brain, within the consent boundary.

§ 23

Compliance, ethics & the central tension

Compliant by design, in every direction — legal and data protection, architectural, security, operational, ethical, reputational. And underneath compliance sits one tension that governs the whole product's conscience.

The compliance posture

- **Age and consent are first-class primitives** that open legal capability. We build the doors the law provides and never chase gaps — intent and documentation are what create violations, so we design to be defensible, not merely undetected.
- **PII is vaulted and segregated** from pseudonymised behavioural data; the two are joined only through governed services.

- **No data is sold**, and private academic data is never used for advertising — ever.

> OPEN — VERIFY BEFORE BUILDING THE CONSENT ARCHITECTURE
>
> India's Digital Personal Data Protection Act children's-data rules (verifiable parental consent, restrictions on tracking and targeted content to minors) must be verified against the **current** published state before the consent architecture and the parent-WhatsApp flow are finalised. They directly shape what the deep engine may do for a minor. This is a build gate in §25 and an open item in §26.

The central tension — named in full

The founding vision documents describe a calm, dependency-reducing academic operating system — a companion that makes a learner more independent and needs them less over time. The growth engine in Part III is habit-forming and aggressive. These genuinely pull against each other, and we do not pretend otherwise. They are reconciled by exactly one rule, stated three times in this document because it is the spine:

> **Engineer compulsion only toward independence.** Every habit mechanic is pointed at mastery, retrieval, independent attempts, and consistency — the behaviours that make the learner need us less. Habit and integrity are made the same loop. Loosen this anywhere and we become the thing we exist to replace.

The Byju's line — what we will not do

- Spike to onboard, **never** to retain — retention is a genuine ritual, not a manufactured anxiety.
- The conversion ask routes to the **parent through pride**, never to the child through guilt.
- The free tier is **genuinely complete** — a real daily win before anything is gated.
- Predicted outcomes are honest ranges, never guarantees; the "capacity" number is real, never manufactured.

Each of those is a place a competitor cuts a corner and a place we will not. Loosen any one and the calm-premium, results-honest brand — the whole differentiator — dies.

§ 24

The conscience layer

Four founding vision documents — covering pedagogy, stakeholder structure, student identity, and feature scope — serve as the pedagogy and ethics conscience of the build. They are never attributed to an individual in any team-facing material. The product adopts the following from them, directly.

The Personal Academic OS

A "Today" spine that surfaces one next best action — what to do now, why, how long, what it builds — rather than a wall of modules.

The assistance ladder

Learn → Coach → Hint → Work-with-me → Check-my-work → Independent. The graduated support Vidya runs, with help that visibly fades toward "let me do it alone."

Academic integrity by design

An AI Contribution Record — what the learner did versus what the AI did is always legible. Integrity is structural, not a policy page.

The Evidence Graph

Every conclusion about a learner links to the evidence behind it; no permanent judgement from a single interaction.

Evidence-weighting

Two 80%s are not equal — the multiplicative mastery model (§07), shown to the learner as plain-language bands.

The ten gap types

A low score is a symptom; the named gap is the diagnosis that drives the intervention (§07).

Source attribution with lineage

Insights carry their provenance and recalculate when permissions change — a data-protection-clean trust layer.

The agent-permission ladder

Recommend → Prepare → Execute-with-permission → Safe-automatic. Communications, purchases, submissions, and deletions always need approval; minors carry parent controls.

Outcomes over screen time

The system will tell a learner to go offline, write on paper, or take a break — it optimises learning, not minutes.

The agentic loop

Observe → Interpret → Plan → Act → Verify → Reflect → Remember — the loop every intelligent action follows.

The academic ontology

"Similarity is not intelligence; relevance is." Retrieval and sequencing are by genuine academic relevance, not surface similarity.

Multimodal & multilingual

Ask or do anything, in any supported form, with natural code-switching across Indian-English, Hinglish, and vernacular.

These are not aspirations bolted on at the end — they are the source the rest of the document answers to. When a growth mechanic and the conscience layer disagree, the conscience layer is consulted, and the reconciliation is always the §23 rule.

PART V

Execution

Not a timeline — a structure. What starts simultaneously, what converges on what, and what gates what. We move by stacking: finish one, move to the next. And the short, honest list of what is still open.

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§ 25

The build structure

Four streams start at once. They converge on a single proof — the atom. Only after the atom is magic do the loops get built, and only after the loops work in one vertical does breadth get poured in. The order is a dependency chain, not a calendar.

The four streams that start simultaneously

FIGURE 4 — FOUR STREAMS → THE ATOM → THE LOOPS → BREADTH

STREAM A · YOURS	**Brand & Experience Language** — taste and decisions; you react to rendered artifacts
STREAM B · DELEGATED	— Claude Code executes; you approve architecture
STREAM C · DELEGATED	— Claude executes; you spot-check quality
GTM · YOURS	**Go-to-market prep** — landing, waitlist, founding status, the story

↓ **converge on** ↓

CONVERGENCE	**THE ATOM** — one middle-school-math topic, transcendent, end to end. The proof gate. Nothing scales until this is magic.

↓

THEN	**THE LOOPS** — session/peak-cut → trigger map → share loops → parent WhatsApp → grant engine + scholarship → friends/levels/XP/trophy → archetype engine → premium + payment

↓

THEN	**BREADTH** — topics → chapters → subjects → grades → competitive exams → create-anything at scale

Stream A — Brand & Experience Language YOUR STREAM

01 Brand strategy & positioning lock — the cognitive-fitness frame, the promise, the voice, the naming system (including the "Superstar" exploration).

02 Visual identity & brandkit — logotype; the colour system (one anchor plus the subject accents); the soothing-premium, playful-like-Brilliant-but-not-kiddie, cool-not-cartoon line; type; motion principles.

03 Signature assets — the knowledge twin as generative hero, the aha sound-and-haptic, the share-artifact templates.

04 The design system as code tokens (v4) — the bridge where brand becomes buildable.

05 Key screens — onboarding, the topic/atom surface, Vidya, the session-cutoff screen, profile and trophy room.

This is where "no compromise on quality, spend whatever is needed" lives — the branding-over-profit investment made concrete.

Stream B — Technical Foundation DELEGATED

01 Repo plus the KGtoPG contract seed — take-and-trim from School (events, mastery, ontology, evaluation, assistance, consent).

02 Identity, auth, and consent primitives — Supabase, phone-OTP, the parent-WhatsApp consent surface.

03 The immutable event lake and outbox — from line one.

04 The model router and AI fabric — the Track-1 gateway, the Track-2 slot, the verification substrate.

05 The three-tier content economy — cache → SLM personalisation → frontier bespoke.

> GATE IN STREAM B
>
> Verify the current DPDP children's-data rules before the consent architecture is finalised (§23, §26).

Stream C — Intelligence Proof & Content DELEGATED

01 The eval-harness spike — free-reasoning grading on messy young-learner math. **The keystone proof, and the first build action.** Not doubt that it is possible — calibration, so the grading agrees with human judgement before any conversion trigger is wired to it.

02 Ontology ingestion of the Indian boards for the launch vertical.

03 The prerequisite graph for that vertical — validated, Khan-skeleton-seeded.

04 The Plexus generation pipeline and the modality library — nuggets, parametric pre-verified simulators.

05 Cache population of verified cores.

Convergence — the atom

One topic in middle-school mathematics, fully realised end to end: the immersive onboarding into it, the orchestrator composing modalities, the build-don't-watch opener, Vidya annotating live with graduated hints and persistent memory, Practice generating clean independent-mastery evidence, the aha, and the chapter-report artifact. This is the proof gate. **Nothing scales until this single topic is transcendent.** This is where "we don't launch broken" is actually enforced — depth-perfect-narrow before wide.

GTM — runs in parallel from early YOUR STREAM

Landing page and waitlist with founding-status and the referral-queue → build-in-public channel setup (its own session) → seed cohort → challenge-loop launch → wave releases tuned to the cost governor.

> The honest shape: A, B, C, and GTM-prep all begin now. Your hands are on A and GTM; the delegated hands are on B and C; the atom is where they converge and where your reaction matters most. You move through the stack by perfecting the atom, then unlocking the loops, then pouring in breadth — never lighting the viral fuse until the atom is magic.

§ 26

The open decisions

Almost everything in this document is locked — the standing rule is that anything not contested is agreed, and it has been applied throughout. These few items are genuinely still open, and they are listed plainly so none is mistaken for settled.

#	OPEN ITEM	WHERE IT RESOLVES
1	**Verify the current DPDP children's-data rules.** The one external fact that gates the consent architecture and the parent-WhatsApp flow.	Stream B gate; do before finalising consent. Can be pulled now.
2	**The premium plan's exact name.** "Superstar" is a strong candidate; the identity-naming principle is locked, the word is not. Test against alternatives that scale 11–17.	Stream A brand work.
3	**Competitive-exam prominence at launch** versus after the core proves.	Sequencing call during the loops.
4	**Exact confirmation of the free / paid boundary** — the dynamic line and which deep capabilities sit behind premium.	Tuned on cohort data once the loops are live.
5	**Final pricing.** ₹999/mo and ₹8,999/yr are now the locked working numbers (anchored to a personal AI tutor, §16). Only fine-tuning remains.	Tuned on cohort data; no longer blocking.
6	**Generosity tolerance for the grant engine** — how freely the discretionary engine grants, by behaviour. A modified "give generously to grow" posture.	Tuned on cohort data; dial in the admin cockpit.

And one item is now **closed** that was open before: pricing is locked to working numbers, and the local AI copycat we briefly weighed as a competitor was assessed and dismissed as noise (§27).

PART VI

The Field

Who else is out there, what is worth taking from each, and why our combination is not replicable by any of them. Recorded so we steal deliberately and differentiate on purpose.

§ 27

The competitive landscape

Most of the products people name as our competitors are playing a different game. The honest read is that no single competitor combines what we do — but several have proven individual mechanics worth taking, and the market's recent history hands us our positioning. We steal deliberately and differentiate on purpose.

What we take — the decisive list

> **From Synthesis Tutor — switch the representation on failure**
>
> The single best mechanic in the field: when a learner stumbles, change the representation and re-approach the concept live (their fractions-as-pizza becomes a number line), rather than only repeating or dropping difficulty. Now a reactive rule in the orchestrator and in Vidya's hint behaviour (§04, §05).

From Khanmigo — the discipline, and the failure mode as a guardrail

Ask before telling (our assistance ladder, validated by the most-cited AI tutor). More valuable is its failure mode: dogmatic Socratic questioning makes learners quit and retreat to passive video. So Vidya guides by default but detects frustration and switches to reveal or work-with-me before disengagement — our pose-struggle-reveal loop already fixes this, and now it is explicit (§04).

From Brilliant — forward selection, and the ceiling that is our wedge

Their "predict the optimal next problem so the learner is ready for a future one" is forward-looking selection, now in our policy (§04). And Brilliant is the closest pedagogical sibling — it validates our entire core thesis (active, pretest-before-teach, scaffolding-fades, spaced retrieval, a daily free meter, gamified status) at ten-million-learner scale. But its content is hand-crafted and slow (it is still building out core courses), STEM-only, English-only, with no India boards, no exam-prep, and no parent. Our generation engine, India-first scope, create-anything door, vernacular, and parent engine are all in that gap. Note the moving target: Brilliant has now shipped an AI tutor of its own, so Vidya's bar has risen — she must be meaningfully deeper (voice, canvas, sees-your-work, persistent memory) than a text-and-visual helper.

From StudyFetch — only the speed of magic

Its core (upload your own materials, get a study toolkit) is the wrong shape for us and its weakness is exactly the thing we make existential — reviewers warn its generated content makes real errors and must be checked. We take one thing: the "something useful in seconds" gratification, pointed at our verified curriculum, plus audio/podcast recaps as a confirmed revision modality. We skip the bring-your-own-content core entirely.

> ### From Physics Wallah and the Byju's wreckage — trust as a feature
>
> Physics Wallah won the post-correction market on affordability, a free public content layer that earns trust before any sale, exam-prep as the core, scholarships, and an authentic founder brand — but it relies on recorded content and doubt resolution, not real-time adaptive AI. We take the free-content-as-trust-builder and the scholarship instinct (§15, §18). Byju's is the anti-pattern bible: it collapsed on predatory sales, EMI loans, and unsustainable acquisition costs, and left Indian parents in a state of fear-of-being-scammed. Our genuinely-free, no-sales-call, cancel-anytime, honestly-billed posture is the wedge that fear creates (§18, §23).

What we explicitly do not take

- **Global competitive leaderboards and leagues** (Brilliant has them) — they breed the exact anxiety we are positioned against; our status system is level-and-trophy, not rank (§14).
- **Any answer-giving or homework-completion behaviour** (Photomath, Socratic-by-Google, generic chatbots) — the opposite of build-don't-watch.
- **The bring-your-own-content cram-tool model** — content-agnostic, pedagogy-light, and correctness-unsafe.

Why the combination is the moat

No competitor is doing the thing we are doing — they each have a slice. Brilliant has the pedagogy but not India, generation, or a deep tutor. Khanmigo has the discipline but is text-only, English-first, and locked to one curriculum. StudyFetch has speed but no correctness or curriculum. Physics Wallah has the brand and reach but not adaptive AI. The defensible position is not any one feature; it is the **combination** — generation-grounded curriculum, a realistic tutor, the evidence-based mastery model, the psychology and habit engine, the create-anything door, the credential endgame, the parent-as-WhatsApp engine, and the cognitive-fitness brand — held together at the ecosystem altitude. That combination is what none of them can assemble quickly, and it is what this entire document describes.

END · CLASSESS LEARNER · MASTER DOCUMENT v2